Code No: 157AC **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Write down the role of algorithms in computing.	[2]
b)	State and explain optimal binary search tree.	[3]
c)	Define greedy algorithm?	[2]
d)	Discuss maximum flow algorithm with an example.	[3]
e)	Define sorting network.	[2]
f)	Explain bio tonic sorting network.	[3]
g)	What is string matching?	[2]
h)	Mention the applications of string-matching algorithms.	[3]
i)	What is vertex cover problem?	[2]
j)	State and explain polynomial time verification.	[3]

PART – B

(50 Marks)

2. Consider the following recurrence T(n)=T(n/3)+T(2n/3)+n. Obtain asymptotic bound using recursion tree method. [10]

OR

3. Given two sequences of characters, P=<MLNOM> Q= <MNOM> Obtain the longest common subsequence. [10]

4. What is an optimal Huffman code for the following set of frequencies based on first 8 Fibonacci numbers: 1,1,2,3,5,8,13,21 [10]

OR

5.a) Define: "topological sort" of a directed acyclic graph. What is the time complexity of topological sort?
b) What is Kruskal's algorithm? Explain the algorithm with a suitable example. [5+5]

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6. Explain in detail about the concept of solving system of linear equations. [10]

OR

7. Give an example to explain in detail about the comparison networks in detail. [10]

8.a) What is the basic idea behind Rabin-Karp algorithm? What is expected running time of this algorithm?

b) Explain in the algorithm for navie bayes string matching? [5+5]

OR

9. Discuss Knuth Morries Pratt (KMP) algorithm. Compute whether the pattern P =10100111 is present in the string T=1001010100111 or not. [10]

10.a) If any NP-complete problem belongs to class P, then is P = NP? Explain.

b) Write a brief note on NP-completeness and the classes-P, NP and NPC. [5+5]

OR

11.a) Define NP complete and NP hard problems with example.

b) How are P and NP problems related? [5+5]

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